# Conditional Synthesis of Training Data for Clinical NLP Tasks in Spanish and English

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**Abstract**

**Background:** Access to clinical narratives for NLP research is severely restricted by privacy regulations, yet these texts are essential for training models for tasks such as ICD-10 coding and clinical named entity recognition (NER). Synthetic data generated by large language models (LLMs) offers a privacy-preserving alternative, but its downstream utility under data constraints and across tasks of varying complexity remains insufficiently characterised.

**Methods:** We propose a conditional generation framework based on instruction-tuning a 20B-parameter LLM, where ICD-10 codes or clinical entities serve as conditioning inputs. We evaluate synthetic corpora across four datasets in English and Spanish covering discharge summaries, referral letters, and radiology reports, assessing downstream utility, corpus fidelity, and memorization-based privacy proxies. A data-scaling ablation varies the generator training fraction (5–100%), and one- and three-shot prompting of the same base model serve as baselines.

**Results:** NER models trained on synthetic data closely approach real-data performance, and even generators trained on a small fraction of real data retain most of the downstream utility. ICD-10 classification achieves comparable overall results, but per-category analysis reveals that rare diagnoses remain poorly captured. N-gram overlap stays low across all conditions, with no verbatim document reproductions found for fine-tuned generators. Fine-tuning outperforms prompting baselines on NER even when trained on 5% of the data, whereas for ICD-10 coding prompting is competitive.
**Conclusions:** Conditional synthesis is a viable strategy for generating clinical NLP training data, particularly for entity-level tasks. Task complexity and label imbalance, rather than data quantity, emerge as the primary determinants of the remaining performance gap.

**Keywords:** Synthetic data, Electronic health records, Large language models, Clinical NLP, ICD-10 coding, Named entity recognition, Privacy

# 1 Introduction

Clinical narratives in electronic health records (EHRs) capture diagnostic reasoning, treatment decisions, and patient context that structured fields cannot represent [1], making them essential for NLP applications such as automated ICD-10 coding [2] and clinical named entity recognition [3]. However, these texts contain sensitive patient information, and regulations such as HIPAA and GDPR severely restrict data sharing across institutions [4]. De-identification offers only partial protection—residual re-identification risks persist and the process itself can degrade data utility [5, 6]. These constraints motivate the development of synthetic alternatives that preserve the statistical and linguistic properties of real clinical text without directly exposing patient records.

Synthetic data generation has emerged as a promising approach to address these challenges, producing artificial clinical text that mimics the statistical and linguistic properties of real EHRs while reducing direct exposure of patient information [7]. Recent advances in large language models (LLMs) have substantially improved the quality and controllability of synthetic text generation. Approaches based on instruction-tuning LLMs with clinical codes or entity descriptions have shown potential for producing diverse and medically coherent notes [8, 9]. Recently, studies have begun exploring the data efficiency of synthetic generation; for instance, Vakili et al. [10] demonstrated that small datasets are sufficient to domain-adapt LLMs for clinical NER when using post-hoc machine annotation. Nevertheless, important gaps remain. Most prior work on synthetic clinical text has focused on ICD-10 coding [7, 9], and although synthetic data has also been used for entity-related tasks [8, 10, 11], these studies typically rely on prompt-based generation, address a single language or dataset, or target a specific application such as de-identification. As a result, the use of conditional generation on NER-annotated corpora, and its systematic comparison against ICD-10 coding under a single evaluation pipeline, is still underexplored. In addition,

the behaviour of conditional generation under explicit data constraints has received little attention, since most prior work fixes a single generator training budget.

In this work, we address these gaps by conducting a comprehensive evaluation of conditional clinical note generation. Our primary objective is to evaluate whether training on LLM-generated clinical text can approach the performance of training on real data across two fundamental Clinical NLP tasks: ICD-10 multi-label classification and clinical named entity recognition (NER). We propose a conditional generation framework based on instruction-tuning, where diagnosis codes or clinical entities serve as conditioning inputs to guide the synthesis of coherent narratives. Building upon recent insights into data-constrained synthesis [10], we conduct a systematic data-scaling analysis to determine the amount of sensitive data required to train an effective conditional generator. Our evaluation spans diverse datasets in both English and Spanish, addressing the critical need for robust multilingual Clinical NLP resources [3].

The contributions of this paper are: (1) a *unified* conditional generation framework that combines instruction-tuning of a Generative LLM with task-specific conditioning—ICD-10 codes for classification, clinical entities for NER—to produce synthetic clinical narratives in English and Spanish across multiple medical domains under a single, consistently applied generation and evaluation protocol; (2) a comprehensive evaluation protocol assessing downstream utility (span-F1, micro-F1, ranking metrics), corpus fidelity, and memorization-based privacy proxies across four clinical datasets spanning two languages, two tasks, and label spaces from 16 chapters to 7,942 codes; (3) a data-scaling ablation revealing that generators trained on as little as 25% of real data retain over 90% of NER utility, while ICD-10 classification exposes a task-complexity-dependent gap that persists even at full data scale; and (4) a comparison with one- and three-shot prompting of the same base model, showing that fine-tuning is essential for NER but yields limited gains for ICD-10 coding.

## 2 Background

The protection of personal health information is recognized as a fundamental aspect of patient rights and is governed by comprehensive regulatory frameworks worldwide [12]. In the United States, the Health Insurance Portability and Accountability Act (HIPAA) defines 18 categories of protected health information that must be removed or masked before EHR data can be used for research purposes [4]. In Europe, the General Data Protection Regulation (GDPR) establishes strict requirements for data minimization and purpose limitation, mandating that personal data be processed only when necessary and stored for limited periods. In Latin America, countries are progressively strengthening their data protection laws; for instance, Chile is currently updating its Personal Data Law to align with international standards and establish dedicated oversight agencies. These regulations create significant barriers to sharing clinical text across institutions, even when the data would benefit medical research and NLP development.

Synthetic data generation offers a mechanism to operationalize these privacy requirements by producing artificial clinical text that preserves the statistical and linguistic properties necessary for model training without directly exposing real patient narratives. Unlike de-identification, which attempts to mask or remove identifiers from existing text, synthetic generation creates entirely new documents that are not derived from any single patient record [7]. This approach can facilitate data sharing, enable reproducible benchmarks, and support method development in settings where access to real EHRs is infeasible. However, it is important to note that synthetic data does not provide formal privacy guarantees; careful evaluation of memorization and leakage risks remains essential [6, 13].

The field of natural language processing underwent a paradigm shift with the introduction of transformer architectures [14]. Transformers leverage self-attention mechanisms to capture long-range dependencies in text, enabling the training of models with hundreds of millions to billions of parameters. Pre-trained language models (PLMs), such as BERT [15], acquire general language understanding through self-supervised objectives on large corpora, which can then be fine-tuned for specific downstream tasks [16]. More recently, instruction-tuning has been established as an effective technique for aligning LLMs with user intent by training on input-output pairs formatted as instructions and responses [17]. This approach enables controlled generation, where the model produces outputs conditioned on specific prompts describing the desired content.

Clinical NLP rapidly adopted these advances, with domain-specific models such as ClinicalBERT [18, 19] demonstrating improved performance on medical language tasks. For Spanish, models like bsc-bio-ehr-es [3] were developed using biomedical and clinical corpora, addressing the scarcity of resources for non-English clinical text. These PLMs have been applied to a wide range of clinical applications, including named entity recognition for diseases, medications, and anatomical structures; relation extraction; and automated coding of diagnoses using classification systems such as ICD-10 [2]. The availability of pre-trained clinical models has accelerated progress, though the need for task-specific training data remains a bottleneck.

The application of LLMs to synthetic clinical text generation has gained considerable momentum as model capabilities have improved. Decoder-only LLMs such as LLaMA and GPT-based models have proven particularly effective for generating clinically coherent synthetic text [7–9]. A key innovation is conditional generation, where structured task labels—such as ICD-10 codes or clinical entity lists—is provided as input to guide the synthesis of coherent narratives. Tang et al. [8] investigated whether LLM-generated synthetic data helps clinical text mining tasks, demonstrating both the promise and the limitations of this approach. Kiefer et al. [7] showed that instruction-tuning LLaMA with ICD-10 code descriptions enables controllable generation with improved diversity. These approaches are particularly valuable in low-resource settings, where synthetic data can augment limited real datasets to improve model training.

Evaluating synthetic clinical text requires assessing its utility for intended downstream applications. The predominant paradigm is to train models exclusively on synthetic data and evaluate them on held-out real datasets [7, 9]. This protocol directly

measures whether synthetic data can serve as a substitute for real training data. Common downstream tasks include multi-label classification for ICD-10 coding, where the model predicts diagnosis codes from clinical notes, and named entity recognition, where the model identifies and classifies clinical entities such as diseases, medications, and body parts. Performance is typically reported using F1 scores at the document level for classification and span-level F1 for NER.

Beyond downstream utility, synthetic data must be evaluated along dimensions of quality and privacy. Fidelity measures assess whether synthetic text resembles real clinical notes in terms of length distributions, vocabulary, and structural properties such as the presence of clinical section headers. Diversity metrics evaluate whether the generated corpus exhibits sufficient lexical and semantic variation, as reduced diversity has been identified as a recurring limitation of synthetic approaches [9, 20]. Privacy assessment typically relies on proxy measures, including exact match rates against training data, n-gram overlap, and near-duplicate detection [7]. While these metrics do not provide formal privacy guarantees, they offer practical indicators of memorization risk. Clinical coherence, assessed through expert review or automated consistency checks, evaluates whether the generated text is medically plausible and internally consistent.

# 3 Data and Methods

Figure 1 illustrates our end-to-end pipeline, which consists of five stages: (1) data preparation, extracting conditioning information (ICD-10 codes or clinical entities); (2) instruction formatting into chat-style templates; (3) supervised fine-tuning of a base LLM via QLoRA; (4) synthetic generation conditioned on held-out task labels; and (5) evaluation across downstream utility, text quality, and privacy proxies.

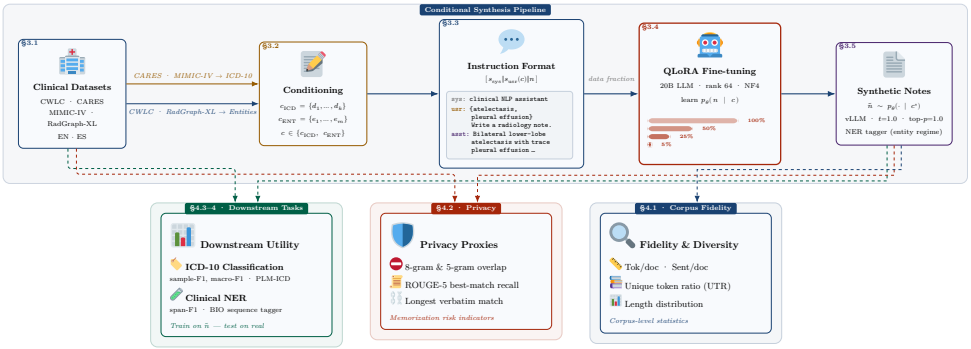



**Fig. 1** End-to-end pipeline for conditional synthetic clinical note generation and evaluation. Stage 1 extracts ICD-10 codes or clinical entities as conditioning inputs; Stage 2 formats each pair into a chat-style instruction template; Stage 3 fine-tunes a 20B-parameter LLM via QLoRA; Stage 4 generates synthetic narratives conditioned on held-out task labels; Stage 5 evaluates the output along three dimensions: downstream task utility, corpus fidelity, and privacy proxies.

## 3.1 Task definition

We formalise synthetic note generation as a conditional text generation task. Let $\mathcal{D} = \{(c_i, n_i)\}_{i=1}^{N}$ denote a training corpus of $N$ conditioning–narrative pairs. A language model $\mathcal{M}_\theta$ is fine-tuned to maximise

$$\mathcal{L}(\theta) = \sum_{i=1}^{N} \log\, p_\theta(n_i \mid c_i), \tag{1}$$

where $c_i$ is the conditioning input and $n_i$ the target narrative. Because the generator is an autoregressive decoder-only LLM, the conditional likelihood of a narrative $n_i = (w_1, \ldots, w_{T_i})$ of $T_i$ tokens factorises as a product of per-token distributions,

$$p_\theta(n_i \mid c_i) = \prod_{t=1}^{T_i} p_\theta(w_t \mid w_{<t},\, c_i)\,, \tag{2}$$

so that maximising Eq. (1) is equivalent to minimising the token-level cross-entropy (next-token prediction loss) over the assistant response, with the loss masked on the system and user (conditioning) tokens. In practice $\theta$ denotes only the QLoRA adapter parameters, as the base weights remain frozen. At inference, a synthetic narrative is sampled token-by-token as $\hat{n} \sim p_\theta(\cdot \mid c)$ using temperature sampling (see Section 3, "Training and generation"). Two conditioning regimes are considered:

**ICD-10 coding regime.** $c = \{(d_1, t_1), \ldots, (d_k, t_k)\}$, where each $d_j$ is a diagnosis code and $t_j$ its textual description. The target $n$ is a discharge summary consistent with those diagnoses.

**Clinical entity regime.** $c = \{(e_1, \ell_1), \ldots, (e_m, \ell_m)\}$, where $e_j$ is an entity mention and $\ell_j \in \mathcal{T}$ its type (e.g., Disease, Medication). The target $n$ is a narrative containing those entities. Both regimes share the same instruction-formatting template.

## 3.2 Datasets

**Table 1** Dataset summary. CWLC and RadGraph-XL: entity regime (NER); CARES and MIMIC-IV: ICD-10 regime (classification).

| Dataset | Conditioning | Lang. | Domain | #Docs | Avg. tok | Labels |
|---|---|---|---|---|---|---|
| CARES | ICD-10 chapters | ES | Discharge | 2,253 | 225 | 16 |
| MIMIC-IV | ICD-10 codes | EN | Discharge | 122,279 | 1,492 | 7,942 |
| MIMIC-S | | | (train) | 13,378 | | |
| MIMIC-M | | | (test) | 19,802 | | |
| CWLC | Entities | ES | Referrals | 9,000 | 34 | 10 types |
| RadGraph-XL | Entities | EN | Radiology | 539 | 447 | 4 types |

We employ four datasets spanning two languages and two tasks, as listed in Table 1. In addition to the following description, examples illustrating each dataset are provided in Figure 2.

**CWLC.** The CWLC corpus [21, 22] consists of 9,000 primary care referral letters in Spanish, collected from the Chilean public health system. Each document is annotated with up to 11 clinical entity types, including Disease, Sign_or_Symptom, Medication, Body_Part, and Abbreviation [23]. The referral letters are short—averaging 34 tokens—and reflect the terse, formulaic writing style of outpatient clinical communication in Latin America. This corpus serves as a low-resource Spanish NER benchmark, where the limited document length and high entity density make it a challenging setting for generative models.

**CARES Corpus.** The CARES corpus [24] comprises 2,253 discharge summaries from a Spanish private medical institution, annotated at the ICD-10 chapter level using a multi-label scheme covering 16 diagnostic categories (e.g., Neoplasms, Respiratory, Digestive). Documents average 225 tokens and exhibit substantial linguistic complexity, including medical abbreviations, drug names, and institutional terminology. CARES is the first publicly released annotated clinical corpus for Spanish ICD-10 coding, and it presents a severe class imbalance: chapter support ranges from 5 to 334 instances, with 4 chapters having fewer than 20 occurrences in the test set.

**MIMIC-IV.** MIMIC-IV [25] is a large-scale English clinical corpus of discharge summaries from the Beth Israel Deaconess Medical Center, linked to fine-grained ICD-10-CM codes. Following the preprocessing and stratified split of Edin et al. [2]—which removes codes with fewer than ten occurrences and ensures class representation across partitions—the 122,279 ICD-10-coded documents (7,942 unique codes) are divided into three subsets: MIMIC-L ($n$=89,098, 72.9%), the large training partition used by Edin et al. for full-scale ICD-10 coding; MIMIC-S ($n$=13,378, 10.9%), a smaller training subset; and MIMIC-M ($n$=19,802, 16.2%), the held-out evaluation set. In our experiments we use MIMIC-S as training data and evaluate all downstream classifiers on MIMIC-M, following the same evaluation protocol of Edin et al. We select MIMIC-S over MIMIC-L because our goal is to assess synthetic data utility under realistic data-constraint conditions, where access to large annotated clinical corpora is limited—a scenario that better represents the target deployment setting for synthetic augmentation. With a median of 1,492 tokens per document, MIMIC-IV constitutes the most demanding generation setting in our study, requiring the model to produce clinically coherent long-form narratives.

**RadGraph-XL.** RadGraph-XL [26] contains 539 English radiology reports annotated with fine-grained entity mentions under a four-class scheme that combines entity type (Anatomy, Observation) with clinical certainty (definitely present, definitely absent, uncertain). Documents average 447 tokens and encompass structured sections typical of radiology reporting. The certainty modifiers introduce significant semantic nuance—distinguishing "lung opacity present" from "lung opacity uncertain"—making this corpus a sensitive test of how well synthetic reports preserve clinical specificity.

Table 2 characterises each real corpus in aggregate, reporting the total and average number of tokens and sentences together with the unique-token ratio (lexical diversity).

The four corpora span more than three orders of magnitude in size, from RadGraph-XL (539 reports, ∼0.22M tokens) to MIMIC-S (13,378 summaries, ∼18M tokens), and in document length, from the terse 34-token referrals of CWLC to the ∼1,350-token discharge summaries of MIMIC-S, providing a broad stress test for conditional generation.

**Table 2** Corpus-level statistics of the four real datasets. Totals are computed over all documents; averages are per document. UTR = unique token ratio (lexical diversity). Token and sentence counts use the same tokeniser as Table 4.

| Dataset | #Docs | Total tok | Avg tok | Total sent | Avg sent | UTR (%) |
|---|---|---|---|---|---|---|
| CWLC | 9,000 | 306K | 34.0 | 17.9K | 1.99 | 10.1 |
| RadGraph-XL | 539 | 221K | 411 | 16.1K | 29.9 | 1.7 |
| CARES | 2,253 | 506K | 225 | 36.5K | 16.2 | 3.4 |
| MIMIC-S | 13,378 | 18.0M | 1,349 | 1.02M | 76.3 | 0.54 |

**RadGraph-XL** *(EN, NER — 447 tok/doc)*
"FINDINGS: The [heart]$_{\text{Anat}^+}$ is [enlarged]$_{\text{Obs}^+}$. Focal [consolidation]$_{\text{Obs}^+}$ in the [left lower lobe]$_{\text{Anat}^+}$. No [pleural effusion]$_{\text{Obs}^-}$. [...]"

**CWLC** *(ES, NER — 34 tok/doc)*
"Pcte con [DM2]$_{\text{Abbr}}$, refiere [cefalea]$_{\text{Sign}}$ intensa y [mareos]$_{\text{Sign}}$. Se solicita [TAC]$_{\text{DiagProc}}$ [cerebral]$_{\text{Body}}$. Tto: [Metformina]$_{\text{Med}}$ 850mg."

**MIMIC-IV** *(EN, ICD-10 — 1,524 tok/doc)*
"DISCHARGE SUMMARY — 72M admitted for acute exacerbation of CHF. PMH: CAD s/p CABG, CKD III, HTN. Hospital course: diuresed with IV furosemide [...]"
Labels: `I50.9` Heart failure , `I25.10` CAD , `N18.3` CKD III , `I10` HTN

**CARES** *(ES, ICD-10 — 225 tok/doc)*
"Pcte masculino 67 años, HTA, DM2, ingresa por disnea progresiva y edema EEII. ECG: FA con RVR. Rx tórax: cardiomegalia, derrame pleural bilateral [...]"
Labels: `Ch. IX` Circulatory , `Ch. IV` Endocrine/Metabolic

**Fig. 2** Illustrative examples from each dataset. Top row: NER tasks (entity spans shown as [text]$_{\text{type}}$; RadGraph-XL superscripts indicate certainty). Bottom row: ICD-10 tasks (document-level labels).

## 3.3 Training and generation

Training examples are converted to a chat-style template: a system message sets the clinical role and language, a user message provides $c$, and the assistant message contains the original note [17]. We use the LLM gpt-oss-20b [27] fine-tuned via QLoRA [28] with 4-bit NF4 quantisation[1]. To study data-scale effects, we train separate generators at 5%, 25%, 50%, and 100% of each training set [10]. For the ICD-10 regime, synthetic notes inherit the original codes as labels; for NER, labels correspond to the conditioning entities that are preserved in the generated text.

**Two-stage protocol and data partitioning.** Our pipeline involves two distinct training stages that never share data with the evaluation set. In *Stage A* (generator

[1]Training acceleration via Unsloth [29], a memory-efficient fine-tuning framework; batch inference via vLLM [30], a high-throughput serving engine for LLMs.

training) the LLM is instruction-tuned on real conditioning–narrative pairs drawn only from the *training* split, using a fraction $f \in \{5, 25, 50, 100\}\%$ of that split. In *Stage B* (downstream training) we take the conditioning inputs $c$ of the *same* training-split documents used in Stage A, sample one synthetic narrative $\hat{n} \sim p_\theta(\cdot \mid c)$ per input, and train the task-specific model (ICD-10 classifier or NER tagger) exclusively on these synthetic pairs $(c, \hat{n})$. The synthetic training set therefore has the same number of documents as the real training fraction it is derived from (a 1:1 replacement), which isolates quality effects from quantity effects. The held-out real test set (MIMIC-M for MIMIC-IV; the official test splits otherwise) is used only for final evaluation and is never seen by the generator or the downstream model. Regarding possible label leakage: the labels used to condition generation in Stage B are training-split labels, which are also the supervision targets for the downstream model. This is intrinsic to using synthetic data as a substitute for the real training set and does not expose any test-set information. The generator's only access to real *text* is through Stage A fine-tuning; the low n-gram overlap reported in Section 4.1 quantifies how little of that text is reproduced verbatim.

**Decoding settings.** One narrative is sampled per conditioning input. For CWLC, CARES, and RadGraph-XL we use temperature = 1.0, top-$p$ = 1.0, and repetition penalty = 1.1, with at most 256, 512, and 500 new tokens, respectively; for MIMIC-S we use temperature = 0.7 and top-$p$ = 0.9 without repetition penalty, with generation bounded by a 4,096-token context. Within each dataset, the configuration is fixed across training fractions and prompting baselines, so that observed differences are attributable to the generator rather than to decoding hyperparameters. Table 3 summarises the fine-tuning and generation configuration.

**Prompting baselines.** To isolate the contribution of fine-tuning, we compare against two prompting baselines that use the same base model without any adapter, the same conditioning inputs, decoding settings, and downstream protocol. In the *one-shot* baseline, the prompt is the fine-tuning template preceded by one real conditioning–narrative pair as a previous user–assistant turn. In the *three-shot* baseline, three real pairs are given together with a single dataset-agnostic instruction to imitate the language, style, formatting, and length of the examples and to include the given entities verbatim. Demonstrations are sampled at random for each input from the training split, excluding the target document and any document in a downstream test split. As with the fine-tuned generators, the note is written directly, without an intermediate reasoning trace.

## 3.4 Evaluation framework

All conditions are evaluated on held-out real test sets, with the task-specific model trained only on synthetic data [7, 9]. For **ICD-10 classification**, we fine-tune PLMs as ICD classifiers [31] and report micro-F1, macro-F1, sample-F1 (CARES), and ranking metrics including precision@$k$, R-precision, and MAP (MIMIC-S). For **clinical NER**, we fine-tune PLMs as BIO sequence-labelling models and report span-level F1.

Span-level F1 for NER follows the standard exact-match convention (a prediction is a true positive only if both the entity boundaries and the type match the gold span). For multi-label ICD-10 coding over a label set $\mathcal{C}$ we report three complementary

**Table 3** Generator fine-tuning (QLoRA/SFT) and synthetic-text generation hyperparameters, shared across all datasets and training fractions, except for decoding, which differs between MIMIC-S and the other corpora.

| Hyperparameter | Value |
|---|---|
| *Base model* | |
| Model | gpt-oss-20b (20B params) |
| Precision / quantisation | bfloat16 / NF4 (4-bit) |
| *LoRA configuration* | |
| LoRA rank $r$ / $\alpha$ / dropout | 64 / 128 / 0.05 |
| Target modules | q_proj, k_proj, v_proj, o_proj |
| Trainable parameters | ∼42M (0.52% of total) |
| *Optimisation* | |
| Optimiser | AdamW |
| Learning rate / schedule | $2 \times 10^{-4}$ / cosine, warmup 0.03 |
| Weight decay | 0.01 |
| Batch (per device × accum. = eff.) | $4 \times 4 = 16$ |
| Epochs / max seq. length | 3 / 2,048 tokens |
| Gradient checkpointing | enabled |
| GPU / acceleration / seed | A100 40GB / Unsloth / 42 |
| *Generation (decoding)* | |
| Strategy | temperature sampling (MIMIC-S: nucleus sampling) |
| Temperature / top-$p$ | 1.0 / 1.0 (MIMIC-S: 0.7 / 0.9) |
| Repetition penalty | 1.1 (MIMIC-S: none) |
| Max new tokens | 256 / 512 / 500 (CWLC / CARES / RadGraph); 4,096-token context (MIMIC-S) |
| Samples per condition | 1 |

aggregations of the per-class F1: *micro*-F1 pools TP/FP/FN counts across all labels and is therefore dominated by frequent chapters; *macro*-F1 is the unweighted mean of per-class F1, which exposes long-tail behaviour; and *sample*-F1 averages the F1 computed per document over the $M$ test documents,

$$F_1^{\text{ma}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F_{1,c}, \qquad F_1^{\text{samp}} = \frac{1}{M} \sum_{j=1}^{M} F_1(y_j, \hat{y}_j). \tag{3}$$

For the large MIMIC-IV label space we additionally report ranking metrics derived from the predicted score vector, namely precision at $k$ ($P$@5 and $P$@8), R-Precision, and mean average precision (MAP $= \frac{1}{M} \sum_{j=1}^{M} \text{AP}_j$, with $\text{AP}_j$ the average precision computed against the gold code set of document $j$), together with the exact match ratio (EMR). Privacy proxies ($n$-gram overlap and ROUGE-5 recall against the closest training document) are computed as in [10, 11] and are indicators of memorisation risk rather than formal guarantees [6].

The chosen PLMs for ICD-10 classification are BiomedBERT [2] (English) and bsc-bio-ehr-es [3] (Spanish), and for clinical NER are RoBERTa [4] (English) and bsc-bio-ehr-es (Spanish). Text quality is assessed via corpus statistics: sentences/document, tokens/document, and unique token ratio [7]. Privacy is measured through 8-gram and

[2] https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract
[3] https://huggingface.co/PlanTL-GOB-ES/bsc-bio-ehr-es
[4] https://huggingface.co/FacebookAI/roberta-base

5-gram overlap, and ROUGE-5 recall against the closest training document [10, 11]. These are proxies for memorization risk, not formal guarantees [6]. Unless otherwise noted, all downstream metrics are reported as mean ± standard deviation over three independent runs; error bars in all figures reflect this variability.

# 4 Results

Results are organised by evaluation dimension: corpus fidelity and privacy, downstream NER, downstream ICD-10 classification, data-scaling analysis, and comparison with prompting baselines.

## 4.1 Corpus fidelity and privacy

Table 4 and Figure 3(a) present corpus fidelity across training fractions. For the shorter-document datasets (CWLC, CARES), document length converges toward the real baseline as the training fraction grows, while RadGraph-XL consistently produces shorter reports. MIMIC-S is a notable exception: synthetic discharge summaries are ∼75% longer than their real counterparts across all fractions, likely because the model generates more verbose clinical narratives when unconstrained by length. For shorter-document corpora, synthetic text exhibits *higher* lexical diversity than real data—contrasting with prior reports of reduced diversity [20] and aligning with Kiefer et al. [7]. This diversity gap narrows monotonically with more training data, suggesting that larger training sets constrain the generator toward more realistic distributions.

**Table 4** Corpus fidelity across training fractions. Δ% indicates relative change vs. real. UTR = unique token ratio (lexical diversity).

| | *Synthetic fraction* | | | | |
|---|---|---|---|---|---|
| Dataset | 5% | 25% | 50% | 100% | Real |
| *Tokens per document* | | | | | |
| CWLC | 42.2 +24% | 35.4 +4% | 32.8 −4% | 34.2 +1% | 34.0 |
| RadGraph | 349 −15% | 357 −13% | 353 −14% | 356 −13% | 411 |
| CARES | 232 +3% | 231 +3% | 237 +5% | 228 +2% | 225 |
| MIMIC-S | 2391 +77% | 2253 +67% | 2327 +72% | 2361 +75% | 1349 |
| *Sentences per document* | | | | | |
| CWLC | 1.66 −17% | 1.54 −23% | 1.55 −22% | 1.56 −22% | 1.99 |
| RadGraph | 14.4 −52% | 13.8 −54% | 14.6 −51% | 13.3 −56% | 29.9 |
| CARES | 7.49 −54% | 8.37 −48% | 7.88 −51% | 7.02 −57% | 16.2 |
| MIMIC-S | 187 +145% | 179 +135% | 196 +157% | 178 +133% | 76.3 |
| *Lexical diversity — UTR (%)* | | | | | |
| CWLC | 24.2 2.4× | 19.6 2.0× | 19.8 2.0× | 18.6 1.9× | 10.1 |
| RadGraph | 9.6 5.7× | 11.2 6.7× | 9.3 5.5× | 8.8 5.2× | 1.7 |
| CARES | 36.0 10.6× | 29.2 8.6× | 27.7 8.1× | 26.3 7.7× | 3.4 |
| MIMIC-S | 0.30 0.6× | 0.29 0.5× | 0.35 0.6× | 0.31 0.6× | 0.54 |

Table 5 and Figure 3(b) present privacy proxy metrics across datasets and training fractions. CARES exhibits the lowest overlap across all conditions, followed by MIMIC-S, RadGraph-XL, and CWLC. The higher CWLC overlap reflects document

brevity: at 34 tokens, standard clinical phrases constitute a large fraction of the n-gram space. MIMIC-S maintains low overlap (5-gram <1.7%) despite producing longer synthetic documents, indicating that the additional text introduces new content rather than memorized passages. For all datasets, overlap increases with training fraction, reflecting the expected trade-off between generation quality and memorization risk. Even at 100%, all values remain low and no exact document matches were found.

**Table 5** Privacy proxy metrics across training fractions. ROUGE-5 = best-match recall against the closest training document.

| | 8-gram ov. (%) | | | | 5-gram ov. (%) | | | | ROUGE-5 rec. | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Dataset | 5 | 25 | 50 | 100 | 5 | 25 | 50 | 100 | 5 | 25 | 50 | 100 |
| CWLC | 1.16 | 1.56 | 1.96 | 2.35 | 3.21 | 4.21 | 4.96 | 5.53 | .132 | .149 | .188 | .232 |
| RadGraph | 0.05 | 0.11 | 0.25 | 0.30 | 0.55 | 0.77 | 1.39 | 1.53 | .008 | .009 | .019 | .019 |
| CARES | 0.00 | 0.05 | 0.03 | 0.12 | 0.01 | 0.15 | 0.15 | 0.30 | .001 | .007 | .006 | .014 |
| MIMIC-S | 0.13 | 0.23 | 0.26 | 0.38 | 1.06 | 1.36 | 1.26 | 1.68 | .010 | .011 | .009 | .017 |

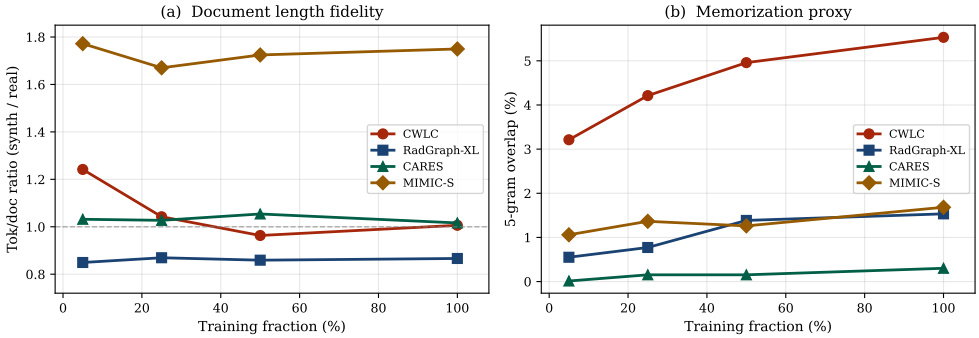



**Fig. 3** Fidelity and privacy metrics across training fractions. (a) Document-length ratio (synthetic / real); dashed line marks parity. (b) 5-gram overlap as a memorization proxy.

## 4.2 Downstream clinical NER

Figure 6(a–b) presents NER downstream performance across training fractions. Both CWLC and RadGraph-XL show monotonically increasing F1 with training fraction. At 100%, CWLC reaches 95% and RadGraph-XL 94% of the respective real upper bounds, indicating that entity-conditioned generation preserves most of the signal needed for span-level recognition.

Figure 4 presents per-entity results comparing real versus synthetic (both at 100%). For CWLC, high-frequency entities (Abbreviation, Disease, Medication) show modest degradation, while lower-frequency entities exhibit larger gaps, revealing frequency-dependent performance loss. Two additional entity types—Anatomy::definitely_absent

($n$=8) and Anatomy::uncertain ($n$=10)—are excluded from this analysis due to insufficient test support. Among the four reported types, Anatomy::definitely_present is well-preserved, but Observation::uncertain shows the largest degradation—a clinically meaningful gap reflecting the difficulty of generating hedged constructions (e.g., "possibly", "cannot exclude") from entity-level conditioning alone.

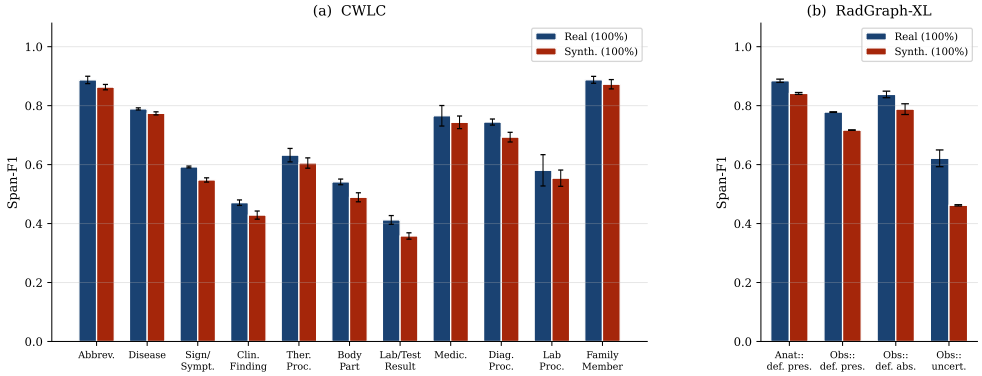



**Fig. 4** Per-entity NER F1 comparing real (100%) vs. synthetic (100%). (a) CWLC (11 entity types, ordered by support). (b) RadGraph-XL (4 entity types).

**Table 6** Downstream results across generator training fractions (%): span-F1 for clinical NER (Figure 6a–b) and sample-F1 for CARES ICD-10 chapter classification (Figure 6c, f). Values are mean ± SD over three runs; "Real" is the upper bound trained on real data.

| | | *Synthetic fraction* | | | | |
|---|---|---|---|---|---|---|
| Task | Dataset | 5% | 25% | 50% | 100% | Real |
| NER (span-F1) | CWLC | 62.4±0.4 | 63.7±1.3 | 65.1±0.7 | 66.7±0.1 | 69.7±0.3 |
| NER (span-F1) | RadGraph-XL | 73.8±0.6 | 75.1±1.0 | 77.7±0.1 | 78.0±0.1 | 79.6±0.0 |
| ICD-10 (sample-F1) | CARES | 52.2±1.5 | 52.3±0.5 | 54.1±2.2 | 60.5±0.6 | 88.9±0.1 |

## 4.3 Downstream ICD-10 classification

Figure 6(c, f) presents CARES ICD-10 classification across training fractions. The best synthetic configuration (100%) reaches ∼93% of real micro-F1 and ∼68% of real sample-F1. Macro-F1—the unweighted average of per-chapter F1 shown in Figure 5—reaches only ∼28% of real, driven by severe degradation on rare chapters. All metrics improve monotonically with training fraction, with the steepest gain between 50% and 100%.

Figure 5 and Table 6 present per-chapter ICD-10 results. All 16 chapters are recognized under synthetic training, but performance varies sharply with label frequency: frequent chapters retain moderate F1 while rare chapters show severe degradation.

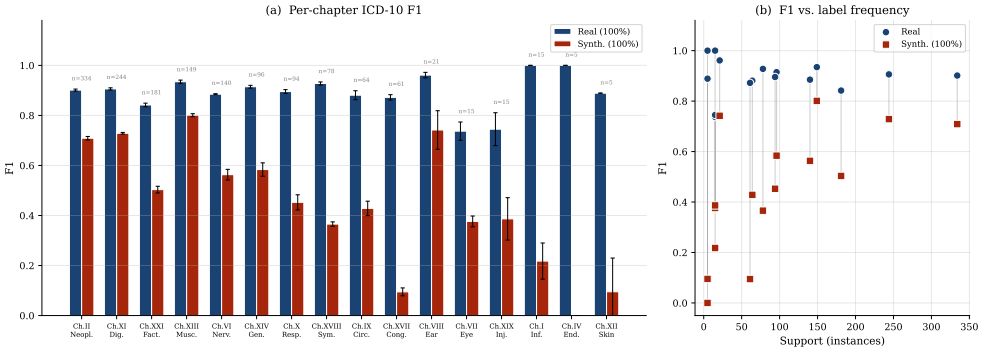



**Fig. 5** ICD-10 classification on CARES. (a) Per-chapter F1 for real (100%) vs. synthetic (100%), ordered by support ($n$ shown above bars). (b) F1 vs. label support; vertical lines connect real–synthetic pairs.

This frequency-dependent pattern is more pronounced than in NER and reflects the multi-label setting, where co-occurrence patterns across chapters must be faithfully reproduced by the generator.

An important caveat concerns the interpretation of Figure 5(b): the support on the horizontal axis is the number of *test-set* instances, and F1 estimated from very few positives is highly variable. The real-data classifier is trained on the full real corpus, so it can still learn a low-frequency chapter well; however, its *measured* F1 at near-zero support is a noisy estimate rather than a reliable performance level. This is evident in the two rarest chapters, both with $n$=5 test instances, whose real F1 differs sharply (Endocrine = 0.89 vs. Skin = 0.33). Consequently, occasional high real-data points at low support in panel (b) reflect estimator variance, and the real–synthetic gap should be read primarily from the higher-support chapters, where the estimates are stable and the degradation of synthetic training is unambiguous.

Table 7 presents MIMIC-IV ICD-10 results. The best synthetic model (100%) reaches 80% of real micro-F1 and 72% of real MAP, confirming that the framework scales to large-vocabulary, fine-grained coding with 7,942 ICD-10-CM codes. Performance improves monotonically across all metrics (Figure 6d, h), with the steepest gains between 25% and 50%. Macro-F1 remains low across all fractions, reflecting the extreme long-tail distribution of ICD-10 codes, where the majority have very few training instances. Notably, synthetic MIMIC-S documents are substantially longer than their real counterparts (∼75% more tokens; Table 4), yet privacy overlap remains low (5-gram <1.7%; Table 5).

## 4.4 Data-scaling analysis

Figure 6 summarizes downstream performance across training fractions for the three datasets with scaling ablations. All three show monotonically increasing performance. For NER, 25–50% of the real training set suffices to achieve over 90% of real-data utility, consistent with Vakili et al. [10]. ICD-10 coding on CARES improves more gradually, reaching ∼68% of real sample-F1 at 100%. MIMIC-S results, reported in Table 7, confirm a similar scaling trend: micro-F1 reaches ∼80% of the real baseline

**Table 7** MIMIC-IV ICD-10 classification results using PLM-ICD across 7,942 codes. Models are trained on real or synthetic versions of MIMIC-S and evaluated on MIMIC-M; the bottom rows correspond to the prompting baselines (Section 4.5). Classification metrics include micro-F1, macro-F1, and exact match ratio (EMR); ranking metrics include precision at $k$ (P@5, P@8), R-Precision, and mean average precision (MAP). Values are percentage.

| | *Classification* | | | *Ranking* | | | |
|---|---|---|---|---|---|---|---|
| Training | F1-Mi | F1-Ma | EMR | P@5 | P@8 | R-Prec | MAP |
| Real (100%) | 53.8±1.2 | 17.1±0.8 | 0.6±0.1 | 72.5±0.4 | 64.0±0.5 | 53.6±0.6 | 56.8±0.5 |
| Synth (5%) | 19.0±0.8 | 0.2±0.1 | 0.0±0.0 | 29.6±0.6 | 25.1±0.5 | 19.3±0.4 | 16.2±0.5 |
| Synth (25%) | 28.4±0.7 | 1.9±0.3 | 0.0±0.0 | 40.1±0.5 | 33.8±0.4 | 26.5±0.5 | 24.3±0.4 |
| Synth (50%) | 35.5±0.6 | 3.9±0.4 | 0.0±0.0 | 53.0±0.4 | 43.9±0.5 | 33.8±0.4 | 33.0±0.5 |
| Synth (100%) | 43.3±0.5 | 7.7±0.5 | 0.1±0.0 | 60.7±0.3 | 51.9±0.4 | 41.1±0.5 | 41.0±0.4 |
| One-shot | 40.1±0.4 | 15.9±1.2 | 0.2±0.0 | 55.8±0.3 | 48.4±0.3 | 39.8±0.4 | 39.4±0.5 |
| Three-shot | 41.2±0.3 | 15.2±0.9 | 0.2±0.0 | 55.8±0.3 | 48.6±0.3 | 40.7±0.1 | 39.8±0.2 |

at full fraction, though macro-F1 remains low due to the extreme label-space size (7,942 codes). The persistent gap across task types suggests that task complexity and label-space size—rather than data quantity alone—are the primary determinants of synthetic data utility.

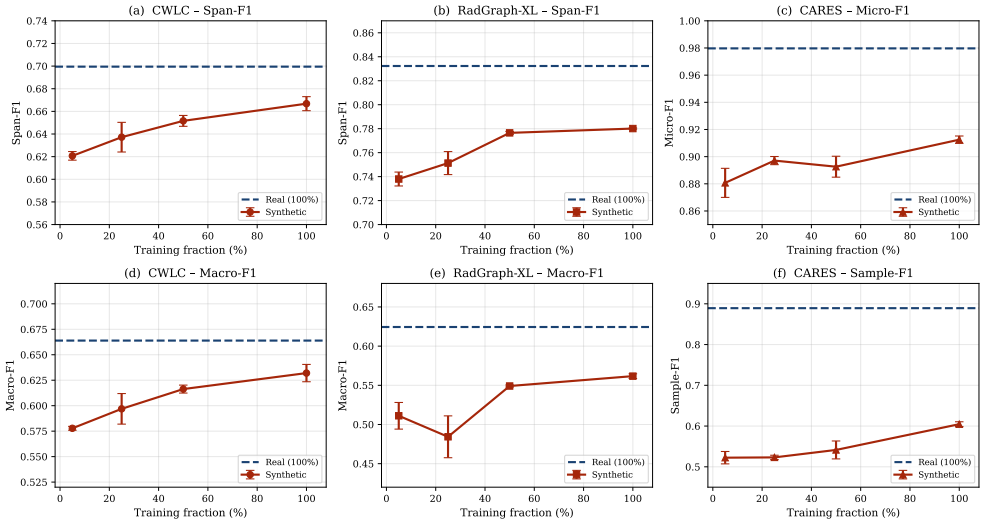



**Fig. 6** Downstream performance as a function of training fraction. Each column corresponds to a dataset. Top row: primary metrics (Span-F1 for NER, Micro-F1 for CARES). Bottom row: secondary metrics (Macro-F1 for NER, Sample-F1 for CARES). Dashed lines indicate the real upper bound. MIMIC-S results are reported separately in Table 7.

## 4.5 Comparison with prompting baselines

Table 8 and the bottom rows of Table 7 compare the fine-tuned generators with the prompting baselines. For NER, fine-tuning is clearly superior: the one-shot baseline reaches 41.1 (CWLC) and 59.1 (RadGraph-XL) span-F1, and the three-shot baseline improves to 52.7 and 69.5, yet both remain below the generator fine-tuned on only 5% of the training data (62.4 and 73.8). For ICD-10 coding the picture differs: on CARES both baselines match the 100% generator in sample-F1 (61.7 and 61.5 vs. 60.5), and on MIMIC-S they reach 40.1–41.2 micro-F1 and 39.4–39.8 MAP, between the 50% and 100% generators, with a higher macro-F1 (15.2–15.9 vs. 7.7).

**Table 8** Downstream results of the prompting baselines compared with fine-tuned generators (5% and 100%) and real data. Values are mean ± SD over three runs (%). MIMIC-S results are reported in Table 7.

| | | *Prompting* | | *Fine-tuned* | | |
|---|---|---|---|---|---|---|
| Dataset | Metric | One-shot | Three-shot | 5% | 100% | Real |
| CWLC | span-F1 | 41.1±1.3 | 52.7±1.5 | 62.4±0.4 | 66.7±0.1 | 69.7±0.3 |
| RadGraph-XL | span-F1 | 59.1±2.9 | 69.5±0.4 | 73.8±0.6 | 78.0±0.1 | 79.6±0.0 |
| CARES | sample-F1 | 61.7±0.7 | 61.5±0.8 | 52.2±1.5 | 60.5±0.6 | 88.9±0.1 |
| CARES | micro-F1 | 90.9±0.5 | 91.4±0.4 | 88.1±1.1 | 91.2±0.3 | 98.0±0.1 |

Table 9 reports fidelity and privacy proxies for the baselines. Without fine-tuning, one-shot outputs deviate most from the real length on the short CWLC referrals (208 vs. 34 tokens), where the model follows a generic clinical-note format instead of the terse referral style; three demonstrations and the style instruction reduce this gap (88 tokens) and bring CARES and RadGraph-XL close to the real length. The 5-gram overlap with the training corpus remains below that of the 100% generators, but in-context examples introduce a different risk: in the three-shot setting, 16 MIMIC-S notes (0.1%) are verbatim copies of one of their demonstrations, and 0.6% reproduce more than 90% of a demonstration's 8-grams.

**Table 9** Fidelity and privacy proxies of the prompting baselines (1s = one-shot, 3s = three-shot) and the 100% fine-tuned generator (FT). Metrics as in Tables 4 and 5.

| | Tokens / doc | | | | 5-gram ov. (%) | | | ROUGE-5 rec. | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Dataset | 1s | 3s | FT | Real | 1s | 3s | FT | 1s | 3s | FT |
| CWLC | 208 | 88 | 34 | 34 | 0.24 | 1.01 | 5.53 | .013 | .041 | .232 |
| RadGraph | 433 | 357 | 356 | 411 | 0.23 | 0.50 | 1.53 | .011 | .012 | .019 |
| CARES | 396 | 216 | 228 | 225 | 0.02 | 0.17 | 0.30 | .002 | .012 | .014 |
| MIMIC-S | 1571 | 1112 | 2361 | 1349 | 0.24 | 1.14 | 1.68 | .005 | .021 | .017 |

# 5 Discussion

Our results demonstrate that conditional synthesis produces training data of sufficient quality to approach real-data performance across both clinical NER and ICD-10 coding tasks, in two languages and across label spaces ranging from 4 entity types to 7,942 ICD-10 codes. For entity recognition, models trained entirely on synthetic text closely match the real upper bound, confirming that entity-level conditioning—where each input mention maps directly to tokens in the generated text—effectively preserves the signal needed for span-level tasks. ICD-10 classification likewise achieves comparable overall performance on both CARES (16 chapters) and MIMIC-S (7,942 codes); however, per-category analysis reveals that rare labels account for most of the remaining gap, consistent with the well-documented long-tail problem in clinical coding [2]. This pattern is more pronounced in MIMIC-S, where the extreme code vocabulary amplifies the challenge: macro-F1 remains low even at 100% training fraction, while micro-F1 and ranking metrics reach ∼80% of the real baseline.

The data-scaling analysis shows that even a small fraction of real data suffices to train generators that retain most downstream utility, with diminishing returns as the fraction grows. This finding is practically significant: it means the framework remains viable in settings where only limited access to sensitive data is available. Synthetic text also exhibits higher lexical diversity than real data—contrasting with prior reports [20] and aligning with Kiefer et al. [7]—and this diversity gap narrows with more training data, suggesting that larger corpora constrain the generator toward more realistic distributions. Our bilingual (English/Spanish) results are consistent with evidence from other languages that synthetic clinical text can support downstream tasks: instruction-tuned or data-constrained generation has been reported for de-identification and NER in Swedish [10], for clinical NER in French [11], and for ICD-coded discharge summaries in English [7, 9]. The task-complexity effect we observe, with entity-level tasks transferring well while long-tail coding lags behind, therefore appears to be a cross-lingual phenomenon rather than a language-specific artefact.

The prompting baselines clarify where fine-tuning matters. Entity-level tasks require the generator to reproduce the terse style of the corpus and to realise the conditioning entities verbatim, which fine-tuning learns from data and a few in-context examples convey only partially. For ICD-10 coding, where the classifier mainly needs the diagnoses to be mentioned in the text, a prompted base model is already competitive, and fine-tuning yields modest (MIMIC-S) or no (CARES) gains. Prompting is not free of privacy risk either: occasional verbatim copies of demonstrations show that in-context examples expose real text more directly than fine-tuning.

Our main experiments fix a 1:1 synthetic-to-real ratio to isolate data quality from quantity, but two observations suggest that generating a synthetic corpus larger than the real data can be useful in specific regimes. First, memorisation proxies remain low and grow only mildly with the training fraction (Table 5), so enlarging the synthetic set does not by itself increase verbatim leakage. Second, because generation is conditioned on the labels, the tail can be conditionally oversampled by drawing more synthetic narratives for underrepresented codes, without touching additional sensitive data.

Finally, while downstream F1 scores measure task utility, they do not capture clinical coherence, factual accuracy, or the plausibility of generated narratives as perceived

by domain experts. Similarly, our privacy assessment relies on n-gram overlap proxies rather than formal guarantees [6]; these indicate low memorization risk but cannot rule out subtler forms of information leakage. Finally, the degradation observed for rare entities and diagnostic chapters points to a need for targeted generation strategies, such as conditional oversampling or retrieval-augmented generation, for tail classes. Our prompting baselines use a small number of randomly sampled demonstrations; retrieving label-similar examples or using more demonstrations may further narrow the gap on NER. Future work should address these gaps through structured clinician evaluation, integration of differential privacy mechanisms, dedicated methods for underrepresented categories, and a systematic study of the synthetic-to-real ratio, including conditional oversampling of the tail.

## 6 Conclusions

This paper presented a conditional generation framework for synthesizing clinical narratives from structured task labels, evaluated across four datasets in English and Spanish covering clinical entity recognition and ICD-10 coding at scales ranging from 16 chapter-level labels to nearly 8,000 fine-grained codes. Synthetic data proves most effective for entity-level tasks, where models trained entirely on generated text closely approach real-data performance even when the generator sees only a fraction of the available data. For diagnostic coding, overall classification remains comparable to the real baseline across both label granularities, yet per-category analysis consistently reveals that rare labels are poorly captured—indicating that task complexity and label imbalance, rather than data quantity, are the primary bottlenecks. Compared with one- and three-shot prompting of the same base model, fine-tuning is essential for entity-level tasks, whereas for ICD-10 coding prompting is already competitive. As conditional synthesis becomes increasingly central to clinical data sharing and NLP development, systematic evaluation of downstream utility, fidelity, and privacy risk is essential to advance the field while safeguarding patient information.

## Availability of data and materials

The CWLC corpus is available at https://zenodo.org/records/7555181. The CARES corpus is available at https://huggingface.co/datasets/chizhikchi/CARES. MIMIC-IV is available through PhysioNet (https://physionet.org/content/mimiciv/). RadGraph-XL is available at https://physionet.org/content/radgraph-xl/1.0.0/. All code for generator fine-tuning, synthetic generation, downstream training, and evaluation, together with the complete hyperparameter configuration (Table 3) and the instruction templates, is publicly available at https://github.com/plncmm/conditional-clinical-synthesis to ensure reproducibility. Synthetic corpora derived from restricted sources (MIMIC-IV) are not redistributed; instead, generation scripts allow their reconstruction by users holding the corresponding data-use agreements.

## Ethics approval and consent to participate

This study uses publicly available clinical datasets. CWLC, CARES, and RadGraph-XL are publicly released corpora. MIMIC-IV requires a data use agreement through PhysioNet. No new data from human subjects were collected, and no additional ethics approval was required.

## Competing interests

The authors declare that they have no competing interests.

## Consent for publication

Not applicable.

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## Authors' contributions

All authors conceived and designed the study. LM led the implementation, performed the experiments, and wrote the initial draft. FV assisted in data preparation and experimental execution. TV contributed to the data-scaling methodology and analysis. JD and CA secured the funding and provided academic oversight. All authors read and approved the final manuscript.

## Acknowledgements

## References

[1] Dalianis, H.: Clinical Text Mining: Secondary Use of Electronic Patient Records. Springer, Cham, Switzerland (2018). https://doi.org/10.1007/978-3-319-78503-5

[2] Edin, J., Junge, A., Havtorn, J.D., Borgholt, L., Maistro, M., Ruotsalo, T., Maaløe, L.: Automated medical coding on MIMIC-III and MIMIC-IV: A critical review and replicability study. In: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 2572–2582. ACM, Taipei, Taiwan (2023). https://doi.org/10.1145/3539618.3591918

[3] Carrino, C.P., Llop, J., Pàmies, M., Gutiérrez-Fandiño, A., Armengol-Estapé, J., Silveira-Ocampo, J., Valencia, A., Gonzalez-Agirre, A., Villegas, M.: Pretrained

biomedical language models for clinical NLP in Spanish. In: Proceedings of the 21st Workshop on Biomedical Language Processing, pp. 193–199. Association for Computational Linguistics, Dublin, Ireland (2022). https://doi.org/10.18653/v1/2022.bionlp-1.19

[4] Yang, X., Lyu, T., Li, Q., Lee, C.-Y., Bian, J., Hogan, W.R., Wu, Y.: A study of deep learning methods for de-identification of clinical notes in cross-institute settings. BMC Medical Informatics and Decision Making **19**(Suppl 5), 232 (2019) https://doi.org/10.1186/s12911-019-0935-4

[5] Bridal, O., Vakili, T., Santini, M.: Cross-clinic de-identification of Swedish electronic health records: Nuances and caveats. In: Proceedings of the Workshop on Ethical and Legal Issues in Human Language Technologies and Multilingual De-Identification of Sensitive Data in Language Resources (LREC 2022), pp. 49–52. European Language Resources Association, Marseille, France (2022)

[6] Carlini, N., Tramèr, F., Wallace, E., Jagielski, M., Herbert-Voss, A., Lee, K., Roberts, A., Brown, T., Song, D., Erlingsson, Ú., Oprea, A., Raffel, C.: Extracting training data from large language models. In: Proceedings of the 30th USENIX Security Symposium, pp. 2633–2650. USENIX Association, Vancouver, BC, Canada (2021)

[7] Kiefer, L., Alabi, J.O., Vakili, T., Dalianis, H., Klakow, D.: Instruction-tuning LLaMA for synthetic medical note generation in Swedish and English. In: Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2025). Association for Computational Linguistics, Varna, Bulgaria (2025)

[8] Tang, R., Han, X., Jiang, X., Hu, X.: Does synthetic data generation of LLMs help clinical text mining? arXiv preprint arXiv:2303.04360 (2023) https://doi.org/10.48550/arXiv.2303.04360

[9] Falis, M., Gema, A.P., Dong, H., Daines, L., Basetti, S., Holder, M., Penfold, R.S., Birch, A., Alex, B.: Can GPT-3.5 generate and code discharge summaries? Journal of the American Medical Informatics Association **31**(10), 2284–2293 (2024) https://doi.org/10.1093/jamia/ocae132

[10] Vakili, T., Henriksson, A., Dalianis, H.: Data-constrained synthesis of training data for de-identification. In: Che, W., Nabende, J., Shutova, E., Pilehvar, M.T. (eds.) Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 27414–27427. Association for Computational Linguistics, Vienna, Austria (2025). https://doi.org/10.18653/v1/2025.acl-long.1329 . https://aclanthology.org/2025.acl-long.1329/

[11] Hiebel, N., Ferret, O., Fort, K., Névéol, A.: Can synthetic text help clinical named entity recognition? a study of electronic health records in French. In: Proceedings

of the 17th Conference of the European Chapter of the Association for Computational Linguistics, pp. 2320–2338. Association for Computational Linguistics, Dubrovnik, Croatia (2023). https://doi.org/10.18653/v1/2023.eacl-main.170

[12] Nampewo, Z., Mike, J.H., Wolff, J.: Respecting, protecting and fulfilling the human right to health. International Journal for Equity in Health **21**(1), 36 (2022) https://doi.org/10.1186/s12939-022-01634-3

[13] Lehman, E., Jain, S., Pichotta, K., Goldberg, Y., Wallace, B.C.: Does BERT pretrained on clinical notes reveal sensitive data? In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pp. 946–959. Association for Computational Linguistics, Online (2021). https://doi.org/10.18653/v1/2021.naacl-main.73

[14] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems, vol. 30. Curran Associates, Inc., Long Beach, CA, USA (2017)

[15] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pp. 4171–4186. Association for Computational Linguistics, Minneapolis, MN, USA (2019). https://doi.org/10.18653/v1/N19-1423

[16] Tunstall, L., Werra, L., Wolf, T.: Natural Language Processing with Transformers: Building Language Applications with Hugging Face. O'Reilly Media, Sebastopol, CA (2022)

[17] Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., Hashimoto, T.B.: Stanford Alpaca: An Instruction-following LLaMA model. https://github.com/tatsu-lab/stanford_alpaca (2023)

[18] Alsentzer, E., Murphy, J.R., Boag, W., Weng, W.-H., Jin, D., Naumann, T., McDermott, M.B.A.: Publicly available clinical BERT embeddings. In: Proceedings of the 2nd Clinical Natural Language Processing Workshop, pp. 72–78. Association for Computational Linguistics, Minneapolis, MN, USA (2019). https://doi.org/10.18653/v1/W19-1909

[19] Huang, K., Altosaar, J., Ranganath, R.: ClinicalBERT: Modeling clinical notes and predicting hospital readmission. arXiv preprint arXiv:1904.05342 (2020)

[20] Libbi, C.A., Trienes, J., Trieschnigg, D., Seifert, C.: Generating synthetic training data for supervised de-identification of electronic health records. Future Internet **13**(5), 136 (2021) https://doi.org/10.3390/fi13050136

[21] Báez, P., Villena, F., Rojas, M., Durán, M., Dunstan, J.: The Chilean waiting list corpus: a new resource for clinical named entity recognition in Spanish. In: Proceedings of the 3rd Clinical Natural Language Processing Workshop, pp. 291–300. Association for Computational Linguistics, Online (2020). https://doi.org/10.18653/v1/2020.clinicalnlp-1.32 . https://aclanthology.org/2020.clinicalnlp-1.32

[22] Báez, P., Villena, F., Zúñiga, K., Jones, N., Fernández, G., Durán, M., Dunstan, J.: Construcción de recursos de texto para la identificación automática de información clínica en narrativas no estructuradas. Revista médica de Chile **149**(7), 1014–1022 (2021) https://doi.org/10.4067/s0034-98872021000701014

[23] Báez, P., Bravo-Marquez, F., Dunstan, J., Rojas, M., Villena, F.: Automatic extraction of nested entities in clinical referrals in Spanish. ACM Transactions on Computing for Healthcare **3**(3), 1–22 (2022) https://doi.org/10.1145/3498324

[24] Chizhikova, M., López-Úbeda, P., Collado-Montañez, J., Martín-Noguerol, T., Díaz-Galiano, M.C., Luna, A., Ureña-López, L.A., Martín-Valdivia, M.T.: CARES: A corpus for classification of Spanish radiological reports. Computers in Biology and Medicine **154**, 106581 (2023) https://doi.org/10.1016/j.compbiomed.2023.106581

[25] Johnson, A.E.W., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., Pollard, T.J., Hao, S., Moody, B., Gow, B., *et al.*: MIMIC-IV, a freely accessible electronic health record dataset. Scientific Data **10**(1), 1 (2023) https://doi.org/10.1038/s41597-022-01899-x

[26] Delbrouck, J.-B., Chambon, P., Chen, Z., Varma, M., Johnston, A., Blankemeier, L., Van Veen, D., Bui, T., Truong, S., Langlotz, C.: RadGraph-XL: A large-scale expert-annotated dataset for entity and relation extraction from radiology reports. In: Findings of the Association for Computational Linguistics: ACL 2024, pp. 12902–12915. Association for Computational Linguistics, Bangkok, Thailand (2024). https://doi.org/10.18653/v1/2024.findings-acl.765

[27] OpenAI: gpt-oss-120b & gpt-oss-20b Model Card (2025). https://arxiv.org/abs/2508.10925

[28] Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L.: QLoRA: Efficient finetuning of quantized LLMs. In: Advances in Neural Information Processing Systems, vol. 36. Curran Associates, Inc., New Orleans, LA, USA (2023)

[29] Han, D., Han, M., Unsloth team: Unsloth. Accessed: 2026-05-31 (2023). https://github.com/unslothai/unsloth

[30] Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C.H., Gonzalez, J.E., Zhang, H., Stoica, I.: Efficient memory management for large language model serving with PagedAttention. In: Proceedings of the 29th Symposium on Operating Systems Principles, pp. 611–626. ACM, Koblenz, Germany (2023). https:

//doi.org/10.1145/3600006.3613165

[31] Huang, C.-W., Tsai, S.-C., Chen, Y.-N.: PLM-ICD: Automatic ICD coding with pretrained language models. In: Proceedings of the 4th Clinical Natural Language Processing Workshop, pp. 10–20. Association for Computational Linguistics, Seattle, WA (2022). https://doi.org/10.18653/v1/2022.clinicalnlp-1.2